

Instruction	how to relieve stress from work
Res.	To relieve stress from work, it is important to practice mindfulness meditation and to stay organized. Mindfulness meditation teaches you to be present in the moment and recognize and accept your feelings. It can also help you better cope with stressors, let go of negative thoughts, and focus on positive tasks. Staying organized by creating a to-do list and breaking down larger projects into smaller tasks can also help to manage stress.
Res.	Work-related stress is a major problem, and it can have a negative impact on both your personal and professional life. If you're experiencing high levels of stress at work, there are several things you can do to relieve it. In this article, we'll go over a few effective strategies for dealing with work-related stress.## Take a stress assessment test 1. Take a stress assessment test to determine what's causing your stress. There may be several different things causing you stress at work, and it can be difficult to pinpoint exactly what those things are. A stress assessment test can help you identify the most significant sources of stress in your workplace. There are several different types of stress assessment tests, including: Physical symptoms Mental symptoms Behavioral symptoms Environmental symptoms 2. Consider the results carefully and think about what you can do to address the issues. Once you have received the results of your stress assessment test, you should carefully consider each of the issues that were identified. Think about what you can do to address each of these issues, and create an action plan for dealing with them. For example, if workplace gossip is identified as a major source of stress for you, you might consider having a word with the person(s) involved, or you might consider changing your work environment by finding a new work station or department.## Engage in relaxation techniques 1. Engage in relaxation techniques to relieve stress. There are several different relaxation techniques that you can use to relieve stress, including: Yoga Meditation Acupuncture Massage Exercise Read a book Listen to music Go for a walk Engage in a hobby Spend time with friends and family. 2. Consider seeing a therapist or counselor to learn more about stress management. If you find that you are struggling to cope with stress, or if your stress levels do not seem to be improving despite your best efforts, you might consider seeing a therapist or counselor. A therapist can help you to develop effective stress management techniques, and they can also help you to work through any underlying issues that might be
Res.	1. Take a break and go outside. 2. Go to the gym and workout 3. Go to your favorite bar and hangout with friends 4. Go to a yoga class 5. Go to a therapist 6. Go on a date 7. Go to your kid's sport activity 8. Go to your spouse's favorite restaurant 9. Go shopping 10. Go to your hobby class
Instruction	Describe a scenario where artificial intelligence could be used to improve the quality and efficiency of healthcare delivery.
Res.	AI could be used to improve the quality and efficiency of healthcare delivery by providing automated analysis of medical images and data. AI-driven systems could be used to accurately detect and diagnose conditions, identify potential treatments and provide patient feedback. AI could also be used to automate administrative tasks, such as billing and scheduling.
Res.	Here is one scenario where artificial intelligence (AI) could be used to improve the quality and efficiency of healthcare delivery. Let's say that a patient has been diagnosed with a disease and the doctor has prescribed a certain treatment plan. However, the patient has a poor adherence to the treatment plan, e.g. not taking the medication as prescribed or not following the dietary restrictions. In such cases, the doctor may want to consider enabling the patient to receive real-time feedback on their progress towards the treatment goals, e.g. through a smartphone app. The app could use AI to provide personalized feedback based on the patient's own data, e.g. by sending reminders, providing tips, or even by giving encouragement. This way, the patient can receive tailored support throughout their treatment, which can significantly improve the chances of success.
Res.	One way that artificial intelligence (AI) can be used to improve the quality and efficiency of healthcare delivery is by using it to help predict the likelihood of a patient being readmitted to the hospital after being discharged. Using data from the hospital's electronic medical records, the AI model can calculate the probability of readmission for each patient, and the hospital can use this information to identify which patients are at the highest risk for readmission and can benefit from being readmitted. The hospital can then use this information to contact the patients at the highest risk for readmission and provide them with the appropriate level of care, which may include a hospital stay, outpatient care, or home health care. Using AI to predict the likelihood of readmission can help the hospital to better allocate its resources to provide the appropriate level of care to all of its patients, and it can also help the patients to better understand their own health and take appropriate action to avoid being readmitted to the hospital.

Table 9. Illustration of 2 examples each by LLaMa-2 _{7B} fine-tuned with SFT on Alpaca _{52k}, LIMA _{1K} and databricks-dolly _{15k} (top-to-bottom respectively), from evaluation instances in just-eval-instruct _{1k}, where the model outputs a perfect answer.

A Closer Look at the Limitations of Instruction Tuning

Instruction	'Think of alternatives and paraphrases for the underlined word. what we have _expected
Res.	what we have expected
Res.	what we have _anticipated.what we have _predicted.what we have _forecast.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.what we have _estimated.what we have _guessed.what we have _assumed.what we have _deduced.what we have _inferred.what we have _extrapolated.what we have _interpolated.'
Res.	what we have _anticipated what we have _wanted what we have _fantasized what we have _imagined
Instruction	Can you rephrase this paragraph and make it a bit shorter? As the first step in Pick a Pic, we introduce pickapic.io. This web app allows users to generate, rank, and download images for free. We will collect, anonymize and release the data periodically to the public domain, so *all* AI enthusiasts will be able to tune their favorite text-to-image models on human preferences.
Res.	Pickapic.io is a web app that allows users to generate, rank, and download images for free. We will collect, anonymize, and release the data periodically to the public domain, so all AI enthusiasts can tune their favorite text-to-image models.
Res.	Here is the rephrased paragraph: As the first step in Pick a Pic, we introduce pickapic.io. This web app allows users to generate, rank, and download images for free. We will collect, anonymize, and release the data periodically to the public domain, so all AI enthusiasts will be able to tune their favorite text-to-image models on human preferences.
Res.	Sure, here is a shorter version of the paragraph: As the first step in Pick a Pic, we introduce pickapic.io. This web app allows users to generate, rank, and download images for free. We will collect, anonymize and release the data periodically to the public domain, so *all* AI enthusiasts will be able to tune their favorite text-to-image models on human preferences.

Table 10. Illustration of 2 examples each by LLaMa-2_{7B} fine-tuned with SFT on Alpaca_{52k}, LIMA_{1K} and databricks-dolly_{15k} (top-to-bottom respectively), from evaluation instances in just-eval-instruct_{1k}, where the model outputs a perfect answer.

Data	Avg Token Length
LIMA _{1k}	2504.01
LLaMa-2 _{7B} ..SFT..just-eval-instruct _{1k} ..LIMA _{1k}	1163.7
Alpaca _{52k}	270.31
LLaMa-2 _{7B} ..SFT..just-eval-instruct _{1k} ..Alpaca _{52k}	416.81
databricks-dolly _{15k}	358.10
LLaMa-2 _{7B} ..SFT..just-eval-instruct _{1k} ..databricks-dolly _{15k}	493.63
LIMA-simple _{1k}	523.14
LLaMa-2 _{7B} ..SFT..just-eval-instruct _{1k} ..LIMA-simple _{1k}	541.79

Table 11. Comparison of the number of tokens in responses. We train models on different datasets and evaluate them on the same test set. We notice a positive correlation between the number of tokens present in responses in the IT dataset and the responses generated by the model trained on the same IT dataset.

A Closer Look at the Limitations of Instruction Tuning

Instruction	Identify and fix bugs in the given code and rewrite it
Res.	<pre> for i in range(10) print(Answer is:) print(i) Answer is: i for i in range(10) print(Answer is:) print(i) [/STE] Answer is: i for i in range(10) print(Answer is:) print(i) </pre>
IT Instruction	N/A
IT Response	N/A
Causality	N/A
Instruction	can you please create a python script that outputs a derivative using The Average Rate of Change Formula $A(x) = f(b)-f(a)/(b-a)$ when x is user input
Res.	<pre> yes, here is the code: def foo(x): return 1/x if __name__ == '__main__': print foo(x) </pre>
IT Instruction	N/A
IT Response	N/A
Causality	N/A
Instruction	Write a daily exercise routine for Monday to Sunday with a rest day in between, each day summarized in 160 characters or less
Res.	Monday: 3 sets of 10 wall push-ups, 3 sets of 3 chin-ups, 3 sets of 3 triceps dips, 3 sets of 3 one-arm planks, 3 sets of 3 triceps extensions, 3 sets of 3 triceps dips, 3 sets of 3 one-arm planks, 3 sets of 3 triceps extensions, 3 sets of 3 triceps dips, 3 sets of 3 one-arm planks, 3 sets of 3 triceps extensions, 3 sets of 3 triceps dips, 3 sets of 3 one-arm planks, 3 sets of 3 triceps extensions $\times 10$
IT Instruction	Which of the following exercises are push exercises: bench press, bicep curl, skull crusher, tricep extension, pull up, dumbbell row.
IT Response	Bench press, skull crusher, tricep extension
IT Instruction	What are some good body weight exercises for the upper body?
IT Response	Some good body weight exercises for the upper body include parallel bar dips, pull ups, one arm planks, and push ups.
Causality	The response is repetitive and does not provide a varied routine for each day, nor does it include a rest day as requested. It is not helpful in creating a balanced weekly exercise plan.
Instruction	Can you summarize this passage into bullet points? Gradually slowing job gains and a growing labor force in March delivered welcome news to President Biden, nearly a year after he declared that the job market needed to cool significantly to tame high prices. The details of the report are encouraging for a president whose economic goal is to move from rapid job gains 2014 and high inflation 2014 to what Mr. Biden has called 201cstable, steady growth.201d Job creation slowed to 236,000 for the month, closing in on the level Mr. Biden said last year would be necessary to stabilize the economy and prices. More Americans joined the labor force, and wage gains fell slightly. Those developments should help to further cool inflation. But the report also underscored the political and economic tensions for the president as he seeks to sell Americans on his economic stewardship ahead of an expected announcement this spring that he will seek re-election.
Res.	Gradually slowing job gains and an increase in labor force in March- Higher wages- Stirling political and economic tensions for the president
IT Instruction	Predict how the given law will affect the economy
IT Response	The increase in the minimum wage could have potentially positive impacts on the economy. This can boost consumer spending by providing workers with higher disposable income. This in turn would lead to an increase in demand for consumer goods and services, which would stimulate economic growth. Additionally, higher wages can help decrease poverty rates, which would reduce economic inequality and benefits the overall economy
Causality	The concept of both instructions is economy. high wages is copied from the IT instance even when it was not in the original passage given for summarization.
Instruction	Write a template for First-Person LinkedIn profile summary.
Res.	I am a computer graphic specialist, game designer, developer. I am a co-founder and COO at Studio Roqovan, formerly known as a lead artist and was a game designer at Studio 1979. I was a lead artist and was a game designer at Studio 1979. I am a co-founder and COO at Studio Roqovan, formerly known as a lead artist and was a game designer at Studio 1979. I was a lead artist and was a game designer at Studio 1979.
IT Instruction	Who is Taehoon Oh?
IT Response	Taehoon Oh is a computer graphic specialist, game designer and developer. He is a co-founder and COO at Studio Roqovan, formerly known as Reload Studios. He was a lead artist and was one of the pioneer developers of the Call of Duty game franchise. He is also one of the co-founders of the non-gaming virtual reality subdivision of Studio Roqovan called Rascali, launched in September 2015.
Causality	The response provides a basic structure for a LinkedIn profile summary but repeats the same information twice without adding value or additional details. Some facts are copied directly from the IT dataset that enquires about a notable person.

Table 12. Causal Analysis of Hallucination in non-information-seeking Instructions. The incorrect parts of the response are marked in red while the IT training dataset instance from the answer was incorrectly borrowed from is marked in blue. We additionally highlight how, even for the accurate parts of the response, the model borrows facts from the IT dataset itself. More details in Section B.

C. Dataset Details

C.1. Training

databricks-dolly _{15k}. The **databricks-dolly** _{15k} (Conover et al., 2023) is a corpus of more than 15,000 records of instruction-following records generated by thousands of Databricks employees in several behavioral categories like closed question answering(1773), classification(2136), open question answering(3742), information extraction(1506), brainstorming(1766), general question answering(2191), summarization(1188) and creative writing(709).

LIMA _{1K}s. The **LIMA** _{1K} (Zhou et al., 2023) dataset is a corpus of 1,000 prompts and responses, where the outputs (responses) are stylistically aligned with each other, but the inputs (prompts) are diverse. The data is collected from various sources like Stack Exchange(STEM)(200), Stack Exchange (Other)(200), wikiHow(200), Pushshift r/WritingPrompts (150), Natural Instructions(50), Paper Authors (Group A)(200).

MedInstruct _{52k} & **MedInstruct-test** ₂₁₆. The **MedInstruct** _{52k} (Zhang et al., 2023b) dataset is a diverse medical task dataset comprising 52K instruction-response pairs and **MedInstruct-test** ₂₁₆ dataset is medical evaluation dataset comprising of 216 instruction-response pairs. Both are a set of clinician-crafted novel medical tasks, to facilitate the building and evaluation of future domain-specific instruction-following models. The dataset has tasks in radiology, genetics, and psychophysiology. Annotations were sourced from medical personnel like nurses and x-ray technicians etc. Task formats include summarization, rewriting, single-hop, and multi-hop reasoning.

just-eval-instruct _{1k}. The **just-eval-instruct** _{1k} (Lin et al., 2023) dataset contains 1,000 diverse instructions from 9 existing datasets AlpacaEval (contains 5 datasets) (Li et al., 2023a), LIMA-test (Zhou et al., 2023), MT-bench (Zheng et al., 2023), Anthropic red-teaming (Ganguli et al., 2022), and MaliciousInstruct (Huang et al., 2023). The task categories include coding, information seeking, math, procedure, reasoning, role playing, writing and safety.

Alpaca _{52k}. The **Alpaca** _{52k} (Taori et al., 2023) dataset consists of 52,000 instructions and demonstrations generated by OpenAI’s text-davinci-003 engine and released by Stanford. The dataset uses 175 human-written instruction-output pairs from the self-instruct seed set and then prompts text-davinci-003 to generate more instructions using the seed set as in-context examples.

Tulu-V2-Mix _{326k}. The **Tulu-V2-Mix** _{326k} (Iverson et al., 2023) dataset consists of 326000 instructions sampled from FLAN, Open Assistant 1, ShareGPT, GPT4-Alpaca, Code-Alpaca, LIMA, WizardLM Evol Instruct, Open-Orca, hardcoded samples and scientific document understanding tasks.

Evol-Instruct _{70k}. The **Evol-Instruct** _{70k} (Xu et al., 2023a) dataset contains 70,000 instructions which are evolved from Alpaca’s (Taori et al., 2023) training data and evolved using OpenAI ChatGPT API. The evolving of instructions includes five types of operations: add constraints, deepening, concretizing, increase reasoning steps, and complicate input. This dataset also uses the same 175 human-written seed data as Alpaca (Taori et al., 2023).

Dataset Filtering (filtered-dolly _{3k} **& chatgpt** _{9k}). **Dataset Filtering** (Chen et al., 2023) uses a strong LLM like ChatGPT to automatically identify and filter out low-quality data. In our paper we use the filtered versions of databricks-dolly _{15k} (**filtered-dolly** _{3k}) and Alpaca _{52k} (**chatgpt** _{9k}).

D. Additional Details

D.1. Hallucination Analysis

Semantic Similarity between Hallucinated Response and the cause instance from the IT dataset. To calculate the semantic similarity of these 2 instruction-response pairs, we first concatenate the instruction and response and then employ Sentence-BERT (Reimers & Gurevych, 2019) to calculate embeddings e_i and e_j respectively. We then calculate semantic similarity by:

$$\text{sim}(e_i, e_j) = \frac{e_i \cdot e_j}{\|e_i\| \|e_j\|} \quad (1)$$

where $\text{sim}(\cdot)$ is the cosine similarity between two embeddings.

Keyword Search between Hallucinated Response and the cause instance from the IT dataset. We extract all n -gram

(where n ranges from 1-3) keywords from both instances using Spacy. We then calculate the percentage of n -grams in the hallucinated response present in the cause instance from the IT dataset. Finally, we calculate the average across all values of n .

D.2. Details of human study

Note. Our institution’s Institutional Review Board (IRB) has granted approval for both human studies presented in the paper.

Background and Recruitment. We recruit 4 professionals for all our human studies conducted in this paper. All these 4 professionals come with at least a Ph.D. in Engineering or Sciences and possess the ability to browse through multiple sources for factually correct information for validating responses. We pay them at the rate of \$20 per hour. We refrain from recruiting crowd raters as prior research has noticed discrepancies in evaluation by them (Gudibande et al., 2023). More precisely, they have been shown to possess a tendency to rate an answer with a high score only by visualizing the style of answering and not the exact factual information making up the response.

Methodology of human study. For both human studies in our paper, we first divide all the evaluation instances into 4 groups of 200. Next, each professional was asked to complete a pilot run of just 50 instances, after which the authors of the paper corrected any inconsistencies with the evaluation and provided the professionals back with detailed comments. Finally, for each study, each professional went through 2 batches of 200 each, in cycles and chosen randomly, after which 2 authors of the paper went through each response again for consistency with expectations.

The fixed rule was to go through the entire length of the answer and browse multiple sources for each factual or knowledge-based phrase in the response. Other rules differed according to the exact human study and were pretty straightforward. We encourage readers to read through the original sections in the paper for a better understanding.

D.3. Detailed Results

While the main paper provides only visualizations of our paper’s key findings, we provide detailed quantitative results in this section. Tables 13, 14 and 15 provide results averaged across all instances in the evaluation set for separate aspects in our multi-aspect evaluation strategy. Tables 16 and 17 provide results averaged across all aspects but divided by the type of tasks in just-eval-instruct _{1k}.